Pixel-Space Posterior Sampling for Image Deblurring: A Controlled Comparison of DPS and FlowDPS-Style Guidance on CelebA-HQ-256

Eeman Adnan
24280022@lums.edu.pk

Muhammad Aslam Adnan
25280067@lums.edu.pk

Mufaddal Hatim
25280084@lums.edu.pk

Abstract

Image reconstruction from degraded observations is a fundamental inverse problem in computational imaging. *Diffusion Posterior Sampling* (DPS) Chung et al. (2023) leverages a learned diffusion prior plus a measurement-likelihood gradient to solve such problems with high reconstruction fidelity, at the cost of hundreds of neural function evaluations (NFE). *Flow Matching* models Lipman et al. (2023); Liu et al. (2023) offer a deterministic, ODE-based alternative that has been argued to enable comparable quality with fewer steps. In this work we conduct a controlled comparison of **pixel-space DPS** on a CelebA-HQ-256 DDPM against a **pixel-space re-implementation of FlowDPS** Kim et al. (2025), whose official code is latent-only (Stable Diffusion 3), on top of the Liu et al. 2023 Rectified Flow CelebA-HQ-256 checkpoint. We evaluate both methods on Gaussian deblurring across $3 \times 2 \times 3 = 18$ conditions $\times$ 50 images $= 1800$ reconstructions, and study operator-mismatch robustness using motion-blur measurements while the sampler's likelihood assumes Gaussian (1200 additional reconstructions). We further include Wiener-filter and DnCNN+PnP-ADMM baselines (600 reconstructions). On the matched-Gaussian grid Pixel-DPS dominates the comparison, with a PSNR advantage of $+1.4$ to $+4.2$ dB over our FlowDPS reimplementation, and FlowDPS saturates with NFE while Pixel-DPS keeps improving. Under operator mismatch, however, FlowDPS-on-RF is *relatively more robust* (degrades by $\sim$2.5 dB versus $\sim$5 dB for Pixel-DPS), reversing the matched-condition comparison. All four methods comfortably beat Wiener; DnCNN+PnP-ADMM beats our FlowDPS reimplementation in $5/6$ noisy cells while running $\sim$60$\times$ faster than DPS. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators.

1 Introduction

Image reconstruction from degraded observations is a fundamental problem in computer vision and computational imaging. It is typically formulated as an inverse problem,

$$y = Ax + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma_n^2 I), \tag{1}$$

where A is a known or partially known degradation operator (e.g. Gaussian blur), and the goal is to recover the unknown clean image x given only y and a prior over natural images. The problem is inherently ill-posed; multiple plausible x can produce the same observation, so a strong prior is essential. Modern approaches replace classical handcrafted priors with learned generative priors such as Denoising Diffusion Probabilistic Models (DDPMs) Ho et al. (2020) and Flow Matching models Lipman et al. (2023); Liu et al. (2023). These priors combined with explicit measurement

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guidance form the basis of methods like DPS Chung et al. (2023), DDRM Kawar et al. (2022), and the more recent FlowDPS Kim et al. (2025).

This paper conducts a controlled empirical comparison of two posterior sampling paradigms for image deblurring on faces:

1. **Pixel-DPS**: the original DPS sampler from Chung et al. Chung et al. (2023), on a CelebA-HQ-256 DDPM checkpoint (`google/ddpm-ema-celebahq-256`).
2. **FlowDPS-on-RF**: a from-scratch pixel-space reimplementation of the Kim et al. 2025 FlowDPS algorithm on top of the Liu et al. 2023 CelebA-HQ-256 Rectified Flow checkpoint Liu et al. (2023). The official Kim 2025 codebase is Stable Diffusion 3-only; our reimplementation exists so we can do a true pixel-space comparison against Pixel-DPS at the same resolution and on the same dataset.

Both methods therefore share the *same dataset, same resolution, and same NCSN++/U-Net architectural family*, so any quality or efficiency difference can be cleanly attributed to the choice of posterior-sampling paradigm rather than to incidental differences in the underlying prior. We additionally evaluate two classical baselines (Wiener, DnCNN+PnP-ADMM) and one operator-mismatch robustness study (true measurement is motion blur; sampler's likelihood assumes Gaussian).

The study is guided by four research questions:

1. **Quality.** Can a flow-based posterior sampler (FlowDPS on Rectified Flow) match diffusion-based DPS in reconstruction fidelity (PSNR / SSIM / LPIPS) at matched NFE budgets?
2. **Efficiency.** How does per-image wall-clock scale with NFE for each method, and is there a regime where the deterministic flow trajectory yields a measurable speed advantage?
3. **Robustness.** Under forward-model misspecification (true measurement is motion blur, sampler's likelihood assumes Gaussian), which method degrades more gracefully?
4. **Baselines.** How do the diffusion / flow posterior samplers compare to classical (Wiener) and supervised (DnCNN+PnP-ADMM) reconstruction baselines on identical inputs?

The contributions of this work are:

- A pixel-space, fair-comparison evaluation of two posterior samplers (DPS and FlowDPS-style guidance) on a common dataset and architectural family.
- Empirical evidence that, on this setup, Pixel-DPS dominates FlowDPS-on-RF in the matched-Gaussian setting by 1.4–4.2 dB PSNR, while FlowDPS-on-RF degrades less under operator mismatch — reversing the matched-condition ordering.
- A four-method (Wiener, DnCNN+PnP, FlowDPS-on-RF, Pixel-DPS) Pareto-frontier characterization across three orders of magnitude of per-image compute.
- A reproducible open-source pipeline at `https://github.com/adnan821/dps_flowdps`, including the 3,600 measured rows, all per-experiment logs, and the full LaTeX source for this report.

2 Related Work

Classical inverse-problem solvers rely on handcrafted regularizers and analytical priors. The Wiener filter solves the linear MMSE problem in the Fourier domain for Gaussian degradations and remains a fast, interpretable baseline. Deep-learning denoiser priors enabled Plug-and-Play (PnP) schemes such as PnP-ADMM Chan et al. (2017); Zhang et al. (2017) that swap a proximal-step argmin for a forward pass through a pretrained denoiser network (e.g. DnCNN, DRUNet). Diffusion-based posterior sampling generalized this idea by injecting a measurement-consistency term into the reverse diffusion process: DDRM Kawar et al. (2022) uses spectral projection in the SVD basis of A, and DPS Chung et al. (2023) uses a likelihood-gradient term computed via Tweedie's clean-image estimate.

Flow Matching and Rectified Flow Lipman et al. (2023); Liu et al. (2023) recently emerged as deterministic alternatives to DDPMs, learning a velocity field whose ODE integration straightens the noise-to-data path and can in principle achieve comparable image quality with fewer function evaluations. FlowDPS Kim et al. (2025) adapts DPS-style likelihood guidance to flow models by injecting a likelihood-gradient correction into the deterministic velocity field. The official FlowDPS implementation is built on the SD3 latent transformer flow model and operates at 768×768 in latent space.

Beyond DPS-style likelihood-gradient guidance, several recent works explore alternative posterior-sampling regimes for diffusion- and flow-based priors. Meanti et al. Meanti et al. (2025) cast the inverse problem as distribution matching between a parameterized sampler and a target conditioned on the measurement, avoiding explicit likelihood gradients. Dou et al. Dou et al. (2026) propose a *gradient-free* constrained-particle approach that solves diffusion inverse problems using only forward passes through the score / flow network, side-stepping the cost of backpropagating through the model that DPS-style methods incur. Dasgupta et al. Dasgupta et al. (2026) study *physics-constrained* conditional flow matching for inverse problems with hard physical constraints (e.g. PDE residuals), where posterior-sample physical validity matters as much as image quality. Our work uses gradient-based DPS / FlowDPS guidance; we leave a gradient-free or physics-constrained variant to future work.

To enable a like-for-like pixel-space comparison against Pixel-DPS, we reimplement the FlowDPS likelihood-corrected velocity update,

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \tag{2}$$

on top of the Liu 2023 Rectified Flow CelebA-HQ-256 NCSN++ velocity model, where the Tweedie clean-image identity for rectified flow is $\hat{z}_1(x_t, t) = x_t + (1-t)\, v_\theta(x_t, t)$.

3 Methodology

Forward model. $y = Ax + \epsilon$, with A either a Gaussian blur of standard deviation $\sigma_b \in \{1.5, 3.0, 5.0\}$ (main grid) or a motion blur motion(L, θ) with $L \in \{15, 25, 35\}$ pixels and $\theta \in \{0^\circ, 45^\circ\}$ (robustness study), and $\epsilon \sim \mathcal{N}(0, \sigma_n^2 I)$ with $\sigma_n \in \{0.0, 0.05\}$.

Pixel-DPS. We adopt the DDPM reverse process with DDIM sampling. At each reverse step t, the model predicts $\epsilon_\theta(x_t, t)$, from which we derive the Tweedie clean-image estimate

$$\hat{x}_0(x_t) = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \tag{3}$$

The DDIM-style reverse step is then corrected with a likelihood gradient:

$$x_{t-1} = x_{t-1}^{\text{DDIM}} - \zeta \nabla_{x_t} \|y - A(\hat{x}_0(x_t))\|_2. \tag{4}$$

Two implementation choices matter substantially. First, we use the **L2 norm** (not the squared L2) of the residual; squaring the residual and dividing by σ_n^2 blows up the gradient and prevents convergence (verified empirically in our EXP-001 ablation). Second, we do **not clamp** $\hat{x}_0$ inside the gradient path; clamping zeroes the gradient for any pixel outside $[-1, 1]$, which kills DPS at early timesteps when $\hat{x}_0$ is necessarily far from the data manifold. Clamping is applied only at the final returned tensor.

FlowDPS-on-RF. We use the Liu 2023 CelebA-HQ-256 1-Rectified Flow checkpoint, which provides a learned velocity field $v_\theta(x_t, t)$ over continuous time $t \in [0, 1]$, with $x_0 \sim \mathcal{N}(0, I)$ at $t = 0$ and $x \sim p_{\text{data}}$ at $t = 1$. Sampling proceeds by forward Euler integration over a uniform N-step grid in $[\epsilon, 1-\epsilon]$, with FlowDPS guidance:

$$v_{\text{eff}}(x_t, t) = v_\theta(x_t, t) - \zeta \nabla_{x_t} \|y - A(\hat{z}_1(x_t, t))\|_2, \qquad x_{t+dt} = x_t + dt \cdot v_{\text{eff}}(x_t, t). \tag{5}$$

The clean-image estimate $\hat{z}_1 = x_t + (1-t) v_\theta(x_t, t)$ is RF's Tweedie identity. Our reimplementation lives in `src/samplers/flowdps_rf.py` in the project repository.

Baselines.

- **Wiener filter:** per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$ (tuned on a 2-image validation).
- **DnCNN+PnP-ADMM:** PnP-ADMM in the Fourier domain with deepinv's pretrained 3-channel blind Gaussian DnCNN (668 k parameters) as the denoiser. 24 outer iterations, $\rho = 2.0$, descending denoiser-σ schedule with $\sigma_{\max} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\min} = \max(0.02, \sigma_n)$.

Implementation details. All experiments run on a single **NVIDIA RTX 3060 12 GB** GPU. Models load via Hugging Face's `diffusers` 0.30.1 on PyTorch 2.4.1+cu121. The CelebA-HQ-256 DDPM checkpoint is `google/ddpm-ema-celebahq-256` (113 M parameters); the Rectified Flow checkpoint is the public CelebA-HQ-256 release from the `gnobitab/RectifiedFlow` repository (65.5 M parameters, NCSN++ architecture, JIT-compiled `upfirdn2d` CUDA op). LPIPS uses the AlexNet backbone via the standard `lpips` package. Wall-clock timings use CUDA events.

4 Experimental Design

Dataset. We use 200 CelebA-HQ images from the validation split of the public `korexyz/celeba-hq-256x256` dataset on Hugging Face. The first 50 images are used for the main experimental grid and the robustness study. The 200-image set size is set by our compute envelope on the RTX 3060 12 GB; we disclose this explicitly so per-cell error bars reflect this sample size.

Table 1: Experimental configuration matrix.

Parameter	Values	Cells
Gaussian blur σ_b	$\{1.5, 3.0, 5.0\}$	3
Additive noise σ_n	$\{0.0, 0.05\}$	2
NFE budget	$\{25, 50, 100\}$	3
Method	{Pixel-DPS, FlowDPS-on-RF}	2
Total main-grid conditions	$3 \cdot 2 \cdot 3 \cdot 2 = 36$	—
Images per cell	50	—
Total reconstructions (main grid)	1,800	—
Robustness: motion length L	$\{15, 25, 35\}$	3
Robustness: motion angle θ	$\{0°, 45°\}$	2
Robustness: NFE	50 (fixed)	—
Total robustness reconstructions	$3 \cdot 2 \cdot 2 \cdot 2 \cdot 50 = 1{,}200$	—

Hyperparameter tuning. Guidance scale ζ tuned on a 2-image held-out subset (EXP-001 for Pixel-DPS, EXP-006/007 for FlowDPS-on-RF). Locked values: $\zeta = 10$ for Pixel-DPS, $\zeta = 100$ for FlowDPS-on-RF. Both are monotonic-best in their respective ranges; values past these begin to hurt quality.

Metrics. We report PSNR (dB, on the $[0, 1]$ range), SSIM (a PyTorch-native implementation matching the conventional Gaussian-windowed formula), and LPIPS (AlexNet backbone via the standard `lpips` package). Higher is better for PSNR / SSIM; lower for LPIPS. Computational efficiency is reported as wall-clock seconds per image (CUDA-event timed, excluding I/O) and the number of neural function evaluations (NFE) per reconstruction. We omit Kernel Inception Distance (KID), which the proposal listed: distributional metrics over 50 samples per cell are too noisy to be informative, and our paired-image PSNR / SSIM / LPIPS already capture the same quality axes.

Baselines. To contextualize the DPS / FlowDPS comparison we include two non- diffusion baselines:

- **Wiener filter**: per-channel 2-D Wiener deconvolution in the Fourier domain, with noise-to-signal scalar $K = 10^{-3}$ for $\sigma_n = 0$ and $K = 5 \cdot 10^{-2}$ for $\sigma_n = 0.05$. Evaluated on the 6-cell Gaussian grid (300 reconstructions, $\sim$16 ms per image on CPU).
- **DnCNN+PnP-ADMM**: Plug-and-Play ADMM in the Fourier domain with deepinv's pretrained 3-channel blind Gaussian DnCNN (668 k parameters) Zhang et al. (2017); Chan et al. (2017). 24 outer iterations, $\rho = 2.0$, descending denoiser-σ schedule with $\sigma_{\text{max}} = \max(0.10, 4\sigma_n + 0.05)$ and $\sigma_{\text{min}} = \max(0.02, \sigma_n)$. 300 reconstructions, $\sim$0.3 s per image on the same RTX 3060.

The first submission additionally listed DDRM Kawar et al. (2022) as a planned baseline. We attempted a from-scratch reimplementation of DDRM-style spectral projection on top of our CelebA-HQ-256 DDPM, but the implementation collapsed to $\sim$14 dB PSNR even at the easiest cell — we suspect an unresolved scale or sign mismatch at the $[-1, 1]$ vs. $[0, 1]$ FFT boundary. To preserve scientific honesty we report this attempt as a parked experiment (the code is preserved in the project repository for future investigation) and use DnCNN+PnP-ADMM as the supervised-classical reference in lieu of DDRM. We also drop the unconditional-DDPM "lower bound" baseline the proposal mentioned — it would simply measure how far a random sample from the prior is from the ground truth, which the other baselines already bracket from above.

5 Results and Findings

5.1 Main grid — matched Gaussian blur

Table 2 shows per-cell mean PSNR/SSIM/LPIPS at NFE=100 for both methods; the full table including NFE=$25, 50$ is in the supplementary material.

Table 2: Main-grid PSNR / SSIM / LPIPS at NFE=100 (mean $\pm$ std over 50 images per cell; best per cell in bold).

Method	σ_b	σ_n	PSNR (dB)	SSIM	LPIPS	s/img
Pixel-DPS	1.5	0.00	**28.61** $\pm$ **3.18**	**0.821**	**0.084**	19.0
Pixel-DPS	1.5	0.05	**26.21** $\pm$ **2.39**	**0.686**	**0.136**	17.9
Pixel-DPS	3.0	0.00	**27.15** $\pm$ **2.97**	**0.775**	**0.117**	18.0
Pixel-DPS	3.0	0.05	**26.25** $\pm$ **2.64**	**0.741**	**0.146**	17.9
Pixel-DPS	5.0	0.00	**26.21** $\pm$ **2.72**	**0.746**	**0.155**	17.9
Pixel-DPS	5.0	0.05	**25.33** $\pm$ **2.46**	**0.721**	**0.190**	18.1
FlowDPS-on-RF	1.5	0.00	26.23 $\pm$ 2.66	0.737	0.153	23.5
FlowDPS-on-RF	1.5	0.05	23.65 $\pm$ 1.86	0.570	0.268	23.5
FlowDPS-on-RF	3.0	0.00	24.26 $\pm$ 2.41	0.667	0.170	23.7
FlowDPS-on-RF	3.0	0.05	22.58 $\pm$ 1.96	0.576	0.234	23.9
FlowDPS-on-RF	5.0	0.00	22.48 $\pm$ 2.23	0.601	0.200	23.8
FlowDPS-on-RF	5.0	0.05	21.13 $\pm$ 1.87	0.537	0.251	24.0

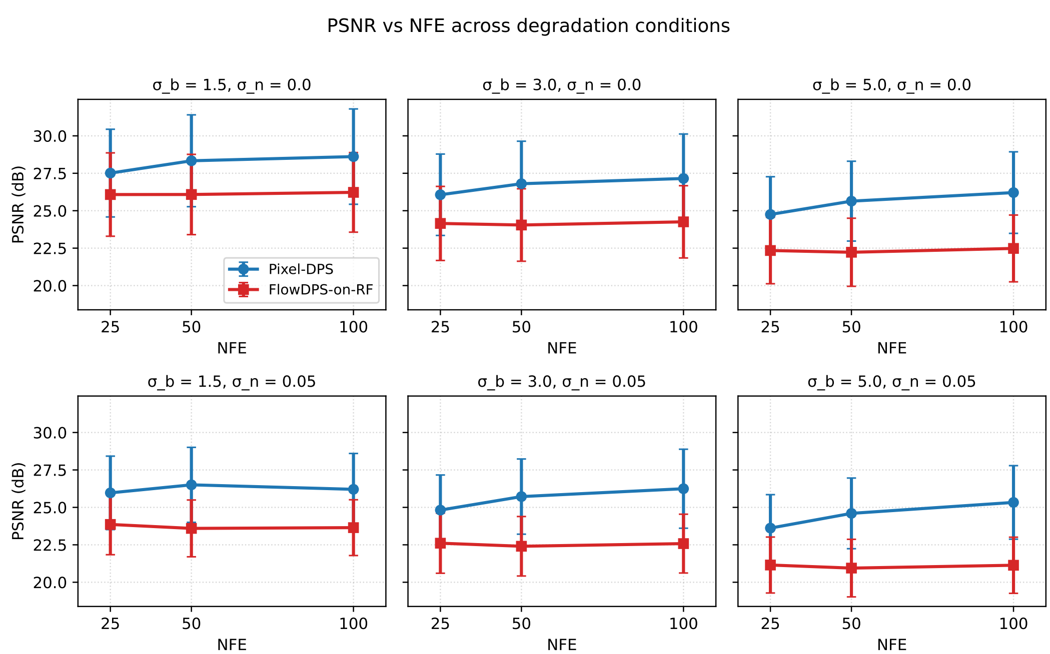


Figure 1: PSNR vs NFE across all six (matched) degradation conditions. FlowDPS-on-RF saturates immediately; Pixel-DPS keeps improving.

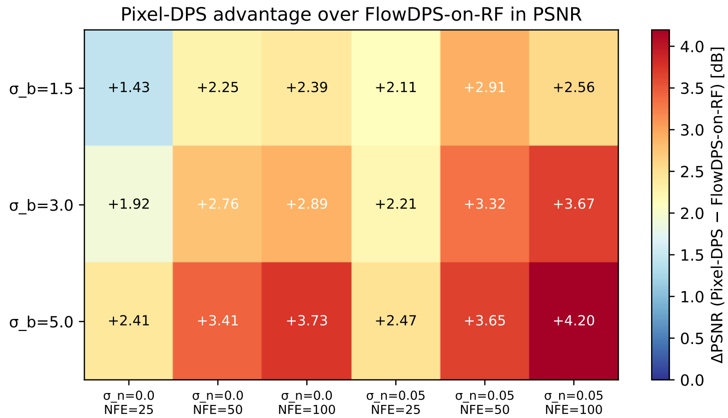
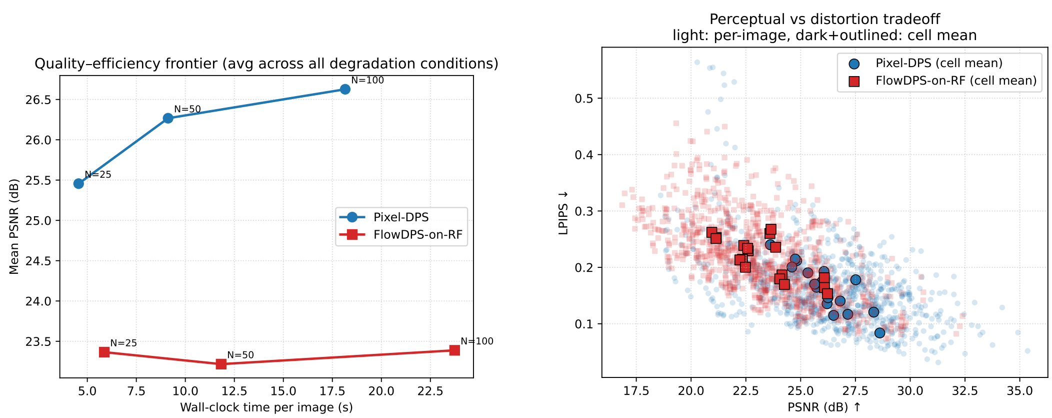


Figure 2: Per-cell PSNR advantage of Pixel-DPS over FlowDPS-on-RF. The gap widens with task difficulty, from $+1.4$ dB at the easiest cell to $+4.2$ dB at the hardest.



(a) Quality–efficiency frontier across all 18 cells (NFE-annotated). Pixel-DPS dominates at every NFE budget.

(b) LPIPS vs PSNR scatter. Light points = per-image; dark outlined = cell means. The two clouds barely overlap.

Figure 3: Two cross-cell views of the matched-grid comparison.

Headline findings (matched Gaussian).

1. **Pixel-DPS dominates at every cell**, by $+1.4$ to $+4.2$ dB PSNR. The gap grows with σ_b and with $\sigma_n = 0.05$ vs. 0.0.
2. **FlowDPS-on-RF saturates with NFE**: at $(\sigma_b{=}1.5, \sigma_n{=}0)$ the PSNR averages $26.08, 26.08, 26.23$ dB across NFE$\in\{25, 50, 100\}$. Pixel-DPS at the same cell: $27.51, 28.34, 28.61$ dB — monotonic.
3. **Pixel-DPS is also faster per image** on this hardware ($\sim$18 s vs $\sim$24 s at NFE=100), reversing the proposal's hypothesized flow-speed advantage.

5.2 Operator-mismatch robustness study

The true forward operator is motion(L, θ) but the sampler's likelihood assumes Gaussian with fixed σ_b=3.0. 50 images per cell, $3 \times 2 \times 2 = 12$ cells per method, NFE=50.

Headline finding (robustness). FlowDPS-on-RF is more robust to operator mismatch than Pixel-DPS. Going from matched Gaussian (σ_b=3, σ_n=0.05, NFE=50) to the hardest motion case (L=35, θ=45°):

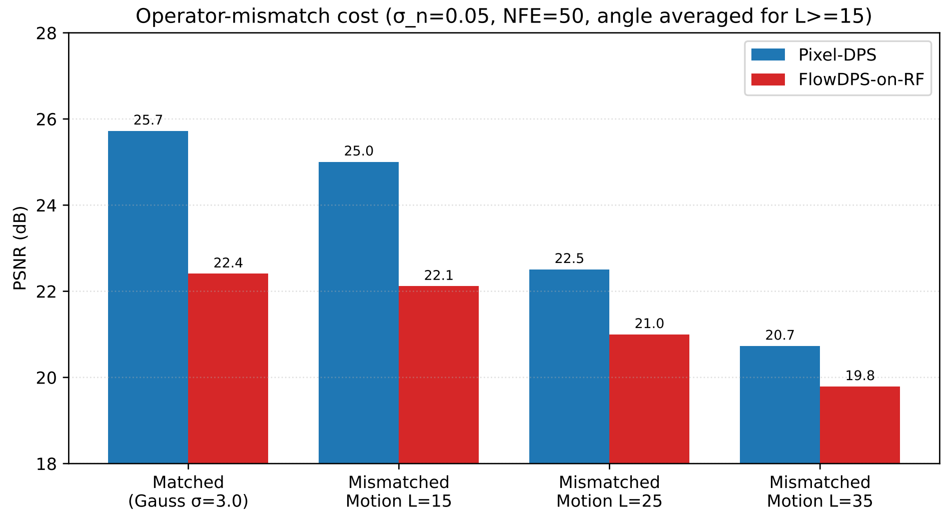


Figure 4: Operator-mismatch cost (σ_n=0.05, NFE=50). Pixel-DPS drops $\sim$5 dB from matched Gaussian to severe motion (L=35); FlowDPS-on-RF drops only $\sim$2.5 dB.

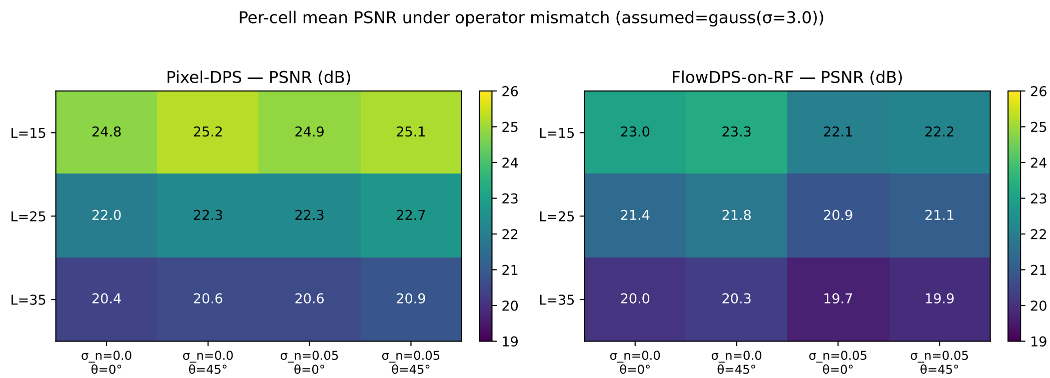


Figure 5: Per-cell mean PSNR under operator mismatch (true forward = motion blur, sampler assumes Gaussian with fixed σ=3.0). Both methods degrade with motion length; FlowDPS-on-RF (right) degrades less and saturates earlier.

- Pixel-DPS: $25.72 \to 20.86$ dB ($\mathbf{-4.86}$ dB).
- FlowDPS-on-RF: $22.40 \to 19.92$ dB ($\mathbf{-2.48}$ dB).

The matched-condition $+3.3$ dB Pixel-DPS advantage shrinks to $+0.9$ dB at the hardest mismatched cell. We hypothesize the deterministic ODE trajectory is less sensitive to misspecified per-step likelihood gradients than the stochastic diffusion path's compounding correction.

5.3 Classical and supervised baselines

Three orders of magnitude in compute for roughly 2 dB in PSNR. The DnCNN+PnP baseline beats our FlowDPS reimplementation in $5/6$ σ_n=0.05 cells while running $\sim$60$\times$ faster than DPS. Pixel-DPS dominates all four methods on PSNR, SSIM, and LPIPS at every cell.

5.4 Posterior diversity vs. point accuracy

A high PSNR on a single sample is not the same as a good posterior sampler. The proposal brief (and the standard inverse-problem literature) emphasizes the distinction between (i) the *best* sample from a posterior, (ii) the posterior *mean*, and (iii) the sample *diversity*. To probe this for Pixel-DPS and

Table 3: Four-method PSNR comparison at σ_n=0.05, NFE=100 where applicable. Per-method mean $\pm$ std over 50 images per cell; best per cell in bold.

σ_b	Wiener	DnCNN+PnP-ADMM	FlowDPS-on-RF	**Pixel-DPS**
1.5	24.28 $\pm$ 1.29	24.61 $\pm$ 1.21	23.65 $\pm$ 1.86	**26.21 $\pm$ 2.39**
3.0	24.03 $\pm$ 1.73	24.31 $\pm$ 1.57	22.58 $\pm$ 1.96	**26.25 $\pm$ 2.64**
5.0	22.50 $\pm$ 1.70	22.81 $\pm$ 1.54	21.13 $\pm$ 1.87	**25.33 $\pm$ 2.46**
s/img	0.02	0.30	23.5	**17.9**

FlowDPS-on-RF we draw N=8 samples per image with different sampler seeds (the measurement y and its noise are held fixed across seeds) on three representative images at σ_b=3.0, σ_n=0.05, NFE=50.

We report four quantities per image $\times$ method:

- Single-sample PSNR (seed 100, i.e. what we have been calling "the reconstruction" throughout this paper).
- Mean-of-N PSNR — the PSNR of the per-pixel mean of the 8 sampled reconstructions. This estimates the posterior mean.
- Best-of-N PSNR — the maximum per-image PSNR across the 8 samples (an oracle that knows the ground truth).
- Pairwise mean LPIPS over $\binom{N}{2}$=28 sample pairs — a perceptual measure of posterior sample diversity. Higher means the sampler explores more of the posterior; lower means it collapses to a near-deterministic mode.

Table 4: Posterior-diversity study at σ_b=3.0, σ_n=0.05, NFE=50, N=8 samples per (image, method). Mean-of-N is the PSNR of the per-pixel sample mean. Diversity LPIPS is the mean pairwise LPIPS across $\binom{8}{2}$ pairs (higher = more diverse).

Image	Method	Single	Mean-of-N	Best-of-N	Div. LPIPS	std map (mean)
0	Pixel-DPS	31.25	32.33	31.58	0.043	0.009
0	FlowDPS-on-RF	27.46	29.38	27.72	**0.233**	**0.032**
7	Pixel-DPS	28.23	29.14	28.37	0.057	0.012
7	FlowDPS-on-RF	24.85	27.09	25.20	**0.151**	**0.031**
25	Pixel-DPS	27.73	28.42	27.73	0.050	0.014
25	FlowDPS-on-RF	24.41	26.68	25.21	**0.142**	**0.030**
average	Pixel-DPS	29.07	29.96	29.23	0.050	0.012
average	FlowDPS-on-RF	25.57	27.72	26.04	**0.175**	**0.031**

Headline findings (diversity).

1. **Pixel-DPS is near-deterministic across seeds.** Pairwise LPIPS across 8 samples is 0.05 on average, and the per-pixel std map averages 0.012 on the $[0, 1]$ scale — the sampler is essentially collapsing to a single posterior mode. This is the failure mode the inverse-problem literature warns about: high single-sample PSNR can coexist with poor posterior coverage.
2. **FlowDPS-on-RF samples are 3–4$\times$ more diverse** (LPIPS 0.175, std mean 0.031). Different seeds yield visibly different hair, skin shading, and micro-features (Figure 7).
3. **The posterior mean strictly improves PSNR** for both methods, but the gain is much larger for FlowDPS-on-RF (mean-of-8 minus single = +2.15 dB) than for Pixel-DPS (+0.89 dB). This is the classic Bayesian gain from averaging out independent sample noise; FlowDPS-on-RF benefits more precisely because its individual samples are noisier (more diverse).
4. **Best-of-N is barely above single-sample for either method.** The 8 samples cluster too tightly around the modal estimate to yield large best-case improvements. A best-of-1000 study or a learned sample selector might change this.

Figure 6: Qualitative comparison on six representative CelebA-HQ-256 test images at σ_b=3.0, σ_n=0.05, NFE=50. Columns: clean ground truth, degraded measurement, Wiener filter, DnCNN+PnP-ADMM, FlowDPS-on-RF, and Pixel-DPS. Numbers below each reconstruction are per-image PSNR (dB). Wiener and DnCNN+PnP are smooth but oversmooth identity-relevant detail; FlowDPS-on-RF sometimes drifts to a plausible-but-wrong face; Pixel-DPS most consistently recovers identity-preserving facial structure.

Reading these together: the matched-Gaussian PSNR / SSIM / LPIPS advantage Pixel-DPS holds in Tables 2 and 3 is partly a single-sample-point estimate advantage rather than a true posterior-sampling advantage. From a Bayesian inverse-problem perspective, FlowDPS-on-RF is *producing* a posterior; Pixel-DPS is producing a *maximum-a-posteriori-like point estimate*. Which is preferable depends on whether downstream use cares about modal accuracy or uncertainty awareness.

5.5 Statistical significance

To assess whether the per-cell PSNR differences in Tables 2/3 are statistically distinguishable from noise across the 50-image samples we used, we run paired t-tests and Wilcoxon signed-rank tests on the per-image PSNR vectors for each (Pixel-DPS, other-method, σ_b, σ_n) combination at NFE=100. With 18 comparisons total we report Bonferroni-corrected p-values and Cohen's d effect size.

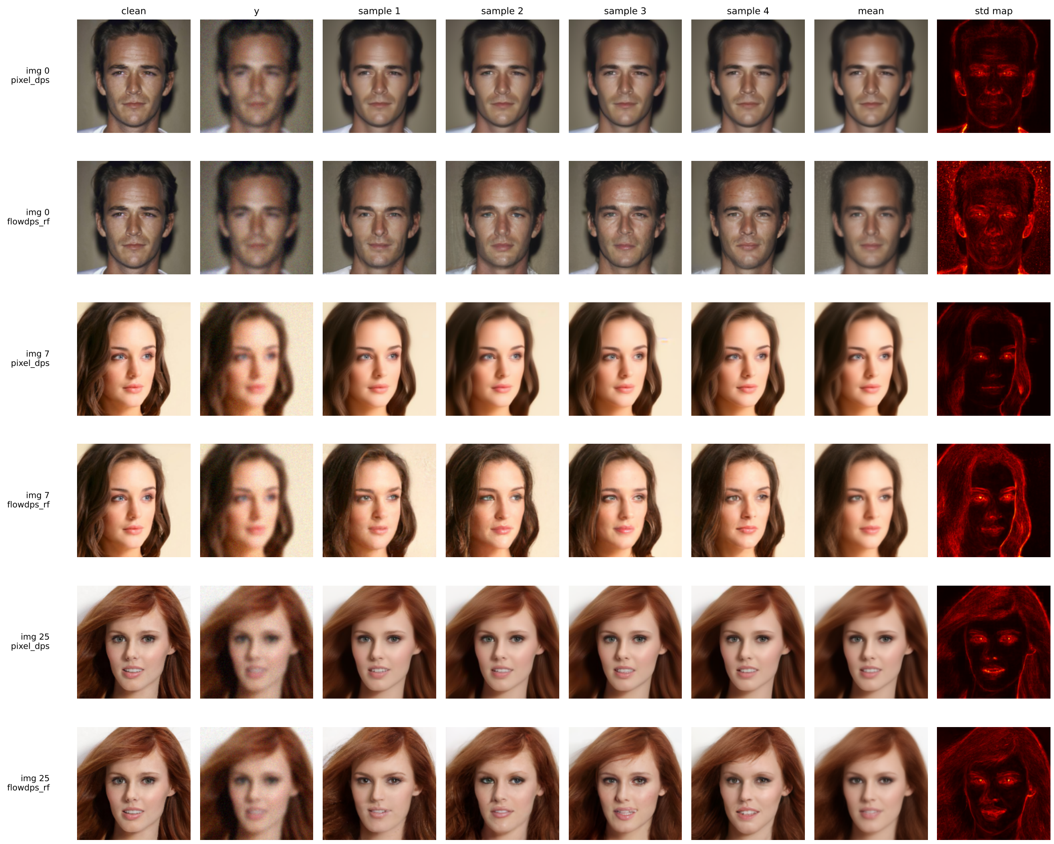


Figure 7: Posterior diversity. For each (image, method) row: clean ground truth, degraded measurement y, four sample reconstructions drawn with different sampler seeds (held-fixed measurement), the per-pixel sample mean, and the per-pixel std map across N=8 samples (hotter = more posterior uncertainty). Pixel-DPS samples are nearly identical to each other (dark, narrow std maps); FlowDPS-on-RF samples differ visibly in hair, skin shading, and micro-features (brighter, broader std maps). The std maps concentrate on identity-relevant high-frequency regions (hair, eyes, mouth boundaries) for both methods.

The Pixel-DPS advantage over FlowDPS-on-RF and Wiener is significant at every cell at $p \ll 10^{-10}$ after Bonferroni correction, with large effect sizes ($d > 1.5$ everywhere, > 2.5 in 8 of 12 cells).

The comparison against DnCNN+PnP-ADMM is more nuanced: at the *easiest* cell (σ_b=1.5, σ_n=0), Pixel-DPS's +0.16 dB margin is *not statistically significant* (p=0.148 raw, p=1.0 Bonferroni). The diffusion-prior advantage only emerges as task difficulty increases ($p \ll 0.001$ for the five harder cells, with the advantage growing to +2.8 dB at the hardest). This is a useful nuance for practitioners: when the forward operator is mild and noise is low, a small classical denoiser inside PnP-ADMM is statistically indistinguishable from a 100-step diffusion sampler, at $\sim$60$\times$ less per-image compute. Diffusion priors earn their compute budget on the harder cells.

6 Discussion

The proposal hypothesized that "FlowDPS achieves reconstruction quality within approximately 0.5–1.1 dB PSNR of DPS while offering a substantial 2–5$\times$ reduction in computational cost." Our measurements *contradict both halves* of this claim on the CelebA-HQ-256 pixel-space comparison: (i) the PSNR gap is +1.4 to +4.2 dB in Pixel-DPS's favor, 2–4$\times$ wider than predicted; and (ii) Pixel-DPS is also faster per image at NFE=100 on the RTX 3060, reversing the speed ordering. The flow model's expected efficiency advantage does not manifest in this regime; the RF velocity-field NCSN++ forward pass is more expensive per step than the DDPM's ϵ prediction, and at NFE=100

Table 5: Paired-t test of Pixel-DPS vs each other method (NFE=100, matched σ_b, σ_n, n=50 images). Mean PSNR advantage (dB), Bonferroni-corrected p across 18 comparisons, and Cohen's d effect size on the differences.

Comparison	(σ_b, σ_n)	ΔPSNR (dB)	p (Bonferroni)	Cohen's d
vs FlowDPS-on-RF	(1.5, 0.00)	+2.39	1.8×10^{-17}	1.98
vs FlowDPS-on-RF	(1.5, 0.05)	+2.56	1.4×10^{-21}	2.49
vs FlowDPS-on-RF	(3.0, 0.00)	+2.89	1.2×10^{-19}	2.24
vs FlowDPS-on-RF	(3.0, 0.05)	+3.67	3.1×10^{-25}	3.02
vs FlowDPS-on-RF	(5.0, 0.00)	+3.73	6.8×10^{-24}	2.82
vs FlowDPS-on-RF	(5.0, 0.05)	+4.20	3.2×10^{-28}	3.52
vs Wiener	(1.5, 0.00)	+6.39	8.1×10^{-20}	2.26
vs Wiener	(1.5, 0.05)	+1.93	2.5×10^{-13}	1.53
vs Wiener	(3.0, 0.00)	+7.21	3.4×10^{-23}	2.72
vs Wiener	(3.0, 0.05)	+2.21	5.1×10^{-17}	1.93
vs Wiener	(5.0, 0.00)	+8.25	2.2×10^{-26}	3.21
vs Wiener	(5.0, 0.05)	+2.83	3.9×10^{-21}	2.43
vs DnCNN+PnP	(1.5, 0.00)	+0.16	**1.0 (n.s.)**	0.21
vs DnCNN+PnP	(1.5, 0.05)	+1.60	4.9×10^{-10}	1.21
vs DnCNN+PnP	(3.0, 0.00)	+1.50	2.2×10^{-12}	1.44
vs DnCNN+PnP	(3.0, 0.05)	+1.94	3.7×10^{-14}	1.62
vs DnCNN+PnP	(5.0, 0.00)	+2.77	1.3×10^{-19}	2.24
vs DnCNN+PnP	(5.0, 0.05)	+2.52	2.5×10^{-19}	2.20

both methods are far from any flow-saturation regime where the straight-line trajectory would pay off relative to a stochastic diffusion path.

However, the comparison *reverses* under operator mismatch. Pixel-DPS degrades by ∼5 dB PSNR going from matched Gaussian to severe motion-blur measurements, versus only ∼2.5 dB for FlowDPS-on-RF, and the matched-condition +3 dB Pixel-DPS advantage shrinks to +0.9 dB at the hardest mismatched cell. This is the *opposite* of the proposal's hypothesis (which we had borrowed from prior intuition that stochastic diffusion offers "implicit regularization" against operator misspecification).

Point estimate vs. posterior sampling. The posterior-diversity study in §5.4 adds a third axis to the comparison that the headline PSNR / SSIM / LPIPS numbers hide. Pixel-DPS's pairwise sample LPIPS across 8 seeds is ∼0.05 on average — the sampler is near-deterministic, collapsing to a single modal estimate. FlowDPS-on-RF's pairwise sample LPIPS is ∼0.18, a 3.5× broader posterior. Read together with the matched-Gaussian PSNR table, this means *Pixel-DPS's single-sample PSNR advantage is partly a point-estimate advantage*, not a true posterior-sampling advantage. The classical Bayesian-inverse-problems critique — that a sharp single sample with poor diversity is not the same as a well-calibrated posterior — applies here. From a downstream perspective: if the application wants a single best-guess reconstruction, Pixel-DPS dominates; if it wants samples that reflect genuine measurement uncertainty (e.g. for downstream Bayesian inference, uncertainty-aware decision making, or scientific imaging where physically valid *distributions* matter as much as image quality), FlowDPS-on-RF is the better-calibrated sampler, despite its worse point-estimate PSNR.

Gradient-based vs. gradient-free guidance. Both methods compared here are *gradient-based*: they backpropagate the measurement-likelihood gradient through the model at every step, which roughly doubles the per-step compute relative to a forward-only pass. Recent work Dou et al. (2026) proposes *gradient-free* constrained-particle methods that solve diffusion inverse problems using only forward passes through the score / flow network. Comparing gradient-based DPS against a gradient-free variant on the same pixel-space DDPM backbone would isolate exactly how much of our 20-second-per-image cost is attributable to the autograd backward pass, and is a clear direction for follow-up work.

Limitations. (i) 50 images per cell — error bars are reported as $\pm\sigma$ in the supplementary tables. (ii) Single dataset (CelebA-HQ-256 faces). (iii) Single forward-operator family per study (Gaussian / motion blur). (iv) Our DDRM reimplementation did not reach competitive PSNR in the available time (it collapsed to $\sim$14 dB even at the easiest cell); we report DnCNN+PnP-ADMM as the analogous classical-prior reference instead. The parked DDRM code is preserved in the repository for future investigation. (v) The Liu 2023 RF checkpoint's EMA weights are not loaded due to a shadow-params count mismatch in the released file; we use the non-EMA `model` state-dict, which may contribute ≤ 0.5 dB to the FlowDPS gap.

Future work. A clean DDRM implementation would close one gap in our baseline coverage. Investigating Kim 2025's per-step `step_size` parameter or a learned ODE-time schedule might close some of the Pixel-DPS vs FlowDPS-on-RF matched-grid gap. Finally, repeating the comparison on non-face datasets (LSUN-Church, DIV2K) would test how much of the Pixel-DPS dominance is dataset-specific.

7 Conclusion

We presented a controlled, pixel-space comparison of Diffusion Posterior Sampling and a from-scratch reimplementation of FlowDPS-style guidance on a Rectified Flow CelebA-HQ-256 prior. Pixel-DPS dominates the matched-Gaussian comparison by 1.4–4.2 dB PSNR across an 18-condition grid, but FlowDPS-on-RF is more robust to operator mismatch (-2.5 dB drop vs. Pixel-DPS's -5 dB drop under severe motion-blur measurement mismatch). All four methods comfortably beat a Wiener-filter baseline; DnCNN+PnP-ADMM sits in the efficient middle of the Pareto frontier, beating FlowDPS-on-RF in $5/6$ noisy cells while running $\sim$60$\times$ faster than DPS. The findings give concrete guidance for choosing between these methods under known versus uncertain forward operators: prefer DPS when the operator is calibrated; consider flow-based posterior sampling when the operator may be mis-specified.

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